

A disproof of Conjecture 00000002913

For every $d \geq 2$ and every $q = 2^k > d^2$, the Keller map $(x^2 - x + y^d, y)$ of degree d is not injective on $\mathbb{F}_q^2$

1 The conjecture

Conjecture 00000002913 reads (English, verbatim):

Definition: finite field this of injection polynomial map: jacobian conjecture of finite field plane. Conjecture: injection spectrum: d -fold polynomial map in $\mathbb{F}_q$ on injection when q greater than d^2 and inverse in q of injection does not exceed of d by Chebyshev lattice gives.

The Chinese text is clearer. It defines the objects as injective polynomial maps of the finite-field plane (“the finite-field version of the Jacobian conjecture”) and conjectures: “a degree- d polynomial map is injective on $\mathbb{F}_q$ when $q > d^2$, and the failure of the converse of injectivity for $q \leq d$ is given by Chebyshev–Lagrange inversion.”

Which clause we refute. The conjecture is a conjunction. We refute its first conjunct, the *threshold clause*: for all d and q with $q > d^2$, every degree- d polynomial map over $\mathbb{F}_q$ is injective. We refute it under each of the following readings of “polynomial map”:

- (R1) any polynomial map $\mathbb{F}_q^2 \rightarrow \mathbb{F}_q^2$ of degree d (no Jacobian hypothesis);
- (R2) a *Keller map* of the plane: a polynomial map $\mathbb{F}_q^2 \rightarrow \mathbb{F}_q^2$ of degree d whose Jacobian determinant is a nonzero constant (the hypothesis of the Jacobian conjecture, which the definition names);
- (R3) a one-variable polynomial map $\mathbb{F}_q \rightarrow \mathbb{F}_q$ of degree d .

Since (R2) is a special case of (R1), one family refutes both. The second conjunct (the “ $q \leq d$ ” clause) is not given a meaning and is not needed: a conjunction with a false conjunct is false.

2 Definitions

A polynomial map of the plane over a field K is a pair $f = (f_0, f_1)$ of polynomials in $K[x_0, x_1]$ (Lean: `PlaneMap K := Fin 2 -> MvPolynomial (Fin 2) K`). Its map on points is $x \mapsto (f_0(x), f_1(x))$ (`PlaneMap.toFun`, evaluation by `MvPolynomial.eval`); its degree is $\max(\deg f_0, \deg f_1)$ with `deg` the total degree (`PlaneMap.degree`, `MvPolynomial.totalDegree`); its Jacobian determinant is $\det(\partial f_i / \partial x_j)$ with Mathlib’s formal partial derivative `MvPolynomial.pderiv` (`PlaneMap.jacobian`). Injectivity is `Function.Injective`. $\mathbb{F}_q$ for $q = p^k$ is Mathlib’s `GaloisField` `p k`, whose cardinality is p^k for $k \neq 0$ (`GaloisField.card`).

3 Proof

For a prime p and $m \geq 1$ let

$$f_{p,m}(x, y) = (x^p - x + y^m, y)$$

(Lean: `asMap p m`).

Lemma 1 (`asMap_not_injective`). *Over every field, $f_{p,m}(0, 0) = (0, 0) = f_{p,m}(1, 0)$; hence $f_{p,m}$ is not injective.*

Proof. $0^p - 0 + 0^m = 0$ and $1^p - 1 + 0^m = 0$ since $p, m \geq 1$. □

Lemma 2 (`asMap_jacobian`). *Over a field of characteristic p , the Jacobian determinant of $f_{p,m}$ is the constant -1 .*

Proof. $\partial_x(x^p - x + y^m) = px^{p-1} - 1 = -1$, $\partial_y(y) = 1$, $\partial_x(y) = 0$, so the determinant is $(-1) \cdot 1 - \partial_y(x^p - x + y^m) \cdot 0 = -1$. □

Lemma 3 (`totalDegree_first, asMap_degree`). *For $p \geq 2$ and $m \geq 1$, $\deg f_{p,m} = \max(p, m)$.*

Proof. The upper bound follows from $\deg(a \pm b) \leq \max(\deg a, \deg b)$. The monomials x^p and y^m both have coefficient 1 (they differ from each other and from x), so the total degree is at least p and at least m . The second component y has degree 1. □

Theorem 4 ((R1); `plain_counterexample`). *For every field K and every $d \geq 2$, the map $(x^2 - x + y^d, y)$ has degree exactly d and is not injective on K^2 . In particular the threshold clause under (R1) fails for every $d \geq 2$ and every finite field $\mathbb{F}_q$ with $q > d^2$, prime fields included.*

Theorem 5 ((R2); `keller_counterexample`). *Over every field of prime characteristic p and for every $m \geq 1$, $f_{p,m}$ has degree $\max(p, m)$, Jacobian determinant -1 , and is not injective.*

Theorem 6 (`conjecture_00000002913_false`). *Let p be prime, $m \geq 1$, and k with $p^k > \max(p, m)^2$. Over $\mathbb{F}_q = \text{GaloisField } p \ k$ one has $q = p^k$, $d := \deg f_{p,m} = \max(p, m)$, $d^2 < q$, Jacobian determinant -1 , and $f_{p,m}$ is not injective on $\mathbb{F}_q^2$.*

Corollary 7 (`keller_counterexample_every_degree, sq_lt_two_pow_two_mul`). *For every $d \geq 2$ and every k with $2^k > d^2$ (for example $k = 2d$), the map $(x^2 - x + y^d, y)$ over $\mathbb{F}_{2^k}$ has degree exactly d , Jacobian determinant -1 , and is not injective. So for every $d \geq 2$ the threshold clause fails at infinitely many $q > d^2$, under (R1) and (R2), and also for odd d (where the characteristic does not divide the degree).*

Theorem 8 ((R3); `univariate_counterexample, univariate_derivative`). *For every field K and $d \geq 2$, $x^d - x \in K[x]$ has degree d and $x \mapsto x^d - x$ is not injective on K (0 and 1 are roots). If $\text{char } K = p$ then $(x^p - x)' = -1$.*

Remark 9 (Context). That $x \mapsto x - x^p$ defeats the naive Jacobian conjecture in characteristic p is classical. The Wikipedia article “Jacobian conjecture” states: “For fields of positive characteristic, the usual Jacobian conjecture is false in one dimension” and, after giving the polynomial $x - x^p$, “There are a few ways to modify the Jacobian conjecture to get around this issue.” The additional term y^m only serves to realize every degree $d \geq 2$ in the plane.

4 Scope and unrefuted readings

Refuted: the threshold clause under (R1) and (R3) for every $d \geq 2$ and every finite field with $q > d^2$; under (R2) for every $d \geq 2$ over $\mathbb{F}_{2^k}$ with $2^k > d^2$, and for every prime p over $\mathbb{F}_{p^k}$ with $p^k > \max(p, m)^2$. Not refuted: (a) the *existential* reading “for $q > d^2$ some degree- d map is injective”, which is true (for instance the triangular automorphism $(x + y^d, y)$); (b) reading (R2) restricted to *prime* fields $\mathbb{F}_p$ (our Keller family needs $q = p^k$ with $k \geq 3$; under (R1) prime fields are covered by `plain_counterexample`); (c) a reading that adds Adjmagbo’s separability hypothesis (p does not divide $[K(x, y) : K(f)]$), under which $f_{p,m}$ is excluded; (d) $d = 1$, where Keller maps are affine bijections and the clause is true; (e) the second conjunct (the “ $q \leq d$ ” / Chebyshev–Lagrange inversion clause), which we do not interpret.

5 Relation to the earlier submission

PR #278 (orionsheep) used the reading (R3) with $x \mapsto x^2$ on $\mathbb{F}_5$ ($5 > 4$, $1^2 = 4^2$). It was merged and then removed in the re-audit of commit 541cf4fb with the reason “residue numerals; no field/map/injectivity objects”: its Lean theorem was a conjunction of statements about natural numbers such as $(4 \cdot 4) \bmod 5 = (1 \cdot 1) \bmod 5$. The present file uses Mathlib’s finite fields `GaloisField p k` with their cardinality, polynomial maps as `MvPolynomial/Polynomial` evaluation, `Function.Injective`, total degree and the formal Jacobian determinant. It refutes the clause for every degree $d \geq 2$ and also under the Jacobian (Keller) hypothesis that the definition names, which the earlier example could not satisfy (the derivative of x^2 is not constant).

6 Lean formalization

File `lean/Conjecture2913/Basic.lean` (Lean 4.33.1, Mathlib), namespace `Tlmc2913`. Main theorems: `conjecture_00000002913_false`, `keller_counterexample_every_degree`, `keller_counterexample`, `plain_counterexample`, `univariate_counterexample`, `univariate_derivative`. `lake build` succeeds; `#print axioms` of each gives `[propxt, Classical.choice, Quot.sound]`. No sorry, no native_decide, no added axioms.